

Sungwon Hwang

STAFF ENGINEER | AI CORE TEAM
Samsung Research | Seoul, South Korea

+82 10-2344-6723 | sungw.hwang@samsung.com | deepshwang.github.io

I am a Staff Engineer at Samsung Research in Seoul, South Korea, where I work on World Action Model to help robots perform a wide range of tasks. I received my Ph.D. in AI from KAIST Graduate School of AI in August 2026, advised by Prof. Jaegul Choo. Previously, I interned at Meta Reality Labs and NAVER LABS.

My research focuses on world models and robot learning, connecting visual understanding and prediction with action in the physical world. I am interested in how generative models and 3D representations can help robots learn transferable skills and adapt to new tasks and environments.

Education

Korea Advanced Institute of Science and Technology

Ph.D | Graduate School of Artificial Intelligence
Advisor: Prof. Jaegul Choo

Seongnam, South Korea
Feb. 2022 - Aug. 2026

Korea Advanced Institute of Science and Technology

M.Sc. | Robotics Program, School of Electrical Engineering
Advisor: Prof. Hyun Myung

Daejeon, South Korea
Feb. 2020 - Feb. 2022

Korea Advanced Institute of Science and Technology

B.Eng. | Department of Mechanical Engineering

Daejeon, South Korea
Aug. 2014 - Feb. 2020

Professional Experience

Samsung Research | AI Core Team

Staff Engineer
Advisor: Jeongseop Kim

Seoul, South Korea
Sept. 2026 - Present

Meta Reality Labs | Codec Avatars Team

Research Intern
Advisor: Timur Bagautdinov and Egor Zakharov

Pittsburgh, USA
May 2025 - Nov. 2025

NAVER LABS | Spatial AI Team

Research Intern
Advisor: Suyong Yeon

Seongnam, South Korea
July 2024 - Oct. 2024

Publications

C: Conference / W: Workshop / J: Journal / P: Preprint / *: Equal contribution

[W01] HumanAnything: Spatially-Aligned Multi-Modal Video Diffusion for Human-Centric Generation

Sungwon Hwang, Egor Zakharov, Junxuan Li, Felix Taubner, Jin Kyu Kim, Itai Druker, Jaegul Choo, Timur Bagautdinov
Proceedings of the European Conference on Computer Vision (ECCV), 2026, MUSTCV Workshop, **Oral presentation**

[C08] SphereDiff: Tuning-free 360° Static and Dynamic Panorama Generation via Spherical Latent Representation

Minho Park*, Taewoong Kang*, Jooyeol Yun, Sungwon Hwang, Jaegul Choo
Association for the Advancement of Artificial Intelligence (AAAI), 2026, **Oral presentation**

Publications (continued)

[C07] *SurFhead: Affine Rig Blending for Geometrically Accurate 2D Gaussian Surfel-based Head Avatars*

Jaeseong Lee*, Taewoong Kang*, Marcel C. Bühler, Min-Jung Kim, **Sungwon Hwang**, Junha Hyung, Hyojin Jang, Jaegul Choo

The International Conference on Learning Representations (ICLR), 2025

[C06] *Effective Rank Analysis and Regularization for Enhanced 3D Gaussian Splatting*

Junha Hyung, Susung Hong, **Sungwon Hwang**, Jaeseong Lee, Jaegul Choo, Jin-Hwa Kim

Advances in Neural Information Processing Systems (NeurIPS), 2024

[C05] *VEGS: View Extrapolation of Urban Scenes in 3D Gaussian Splatting using Learned Priors*

Sungwon Hwang*, Min-Jung Kim*, Taewoong Kang, Jayeon Kang, Jaegul Choo

Proceedings of the European Conference on Computer Vision (ECCV), 2024

[C04] *FaceCLIPNeRF: Text-driven 3D Face Manipulation using Deformable Neural Radiance Fields*

Sungwon Hwang, Junha Hyung, Daejin Kim, Min-Jung Kim, Jaegul Choo

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023

[C03] *Local 3D Editing via 3D Distillation of CLIP Knowledge*

Junha Hyung, **Sungwon Hwang**, Daejin Kim, Hyunji Lee, Jaegul Choo

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023

[C02] *Equivariance-bridged SO(2)-Invariant Representation Learning using Graph Convolutional Network*

Sungwon Hwang, Hyungtae Lim, Hyun Myung

The British Machine Vision Conference (BMVC), 2021

[J01] *ERASOR: Egocentric Ratio of Pseudo Occupancy-based Dynamic Object Removal for Static 3D Point Cloud Map Building*

Hyungtae Lim, **Sungwon Hwang**, Hyun Myung

IEEE Robotics and Automation Letters (RA-L), 2021

[C01] *Normal Distributions Transform is Enough: Real-time 3D Scan Matching for Pose Correction of Mobile Robot under Large Odometry Uncertainties*

Hyungtae Lim*, **Sungwon Hwang***, Sungjae Shin, Hyun Myung

International Conference on Control, Automation and Systems (ICCAS), 2020

[P01] *Text2Control3D: Controllable 3D Avatar Generation in Neural Radiance Fields using Geometry-Guided Text-to-Image Diffusion Model*

Sungwon Hwang, Junha Hyung, Jaegul Choo

arXiv preprint, 2023